Day-Ahead Prosumer Forecasting & Cost-Optimal Procurement Engine

E.ON Hackathon Challenge — Comprehensive Technical Briefing

1. Executive Summary & Problem Formulation

The transition to decentralized renewables introduces compounding volatility for energy suppliers and Distribution System Operators (DSOs). As households electrify with heat pumps and rooftop photovoltaic (PV) systems, grid net load oscillates between weather-driven thermal peaks and mid-day solar supply surges. In European day-ahead power auctions (e.g., EPEX SPOT), market participants clear hourly positions by 12:00 CET on day D−1 for all 24 settlement intervals of day D.

Forecasting errors incur asymmetric penalties on the intraday balancing market: deficits (under-procurement) trigger high replacement penalties (c_short ≈ €35/MWh), whereas surpluses (over-procurement) must be sold at steep discounts (c_long ≈ €15/MWh). We present an end-to-end forecasting and decision engine covering **Levels 0 through 3** that aligns 15-minute smart meter readings with hourly weather forecasts, automatically detects prosumers, quantifies forecast uncertainty, and minimizes portfolio financial exposure.

2. Elegance & Simplicity of Design Choices

- **Strict Temporal Causality (Zero Lookahead):** In day-ahead bidding, smart-meter data from day D−1 afternoon is not yet settled. All autoregressive lag features strictly enforce t ≥ 24 h (lags at t−24 h, t−48 h, t−168 h, and prior-day rolling statistics). Weather signals represent day-ahead numerical weather predictions (NWP).
- **Physics-Informed Thermodynamics:** Heat pumps exhibit non-linear threshold activation. We incorporate Heating Degree Hours, HDH_t = max(0, 15.5 °C − T_t), to capture building heat loss, thermal inertia gradients, and wind-chill convective cooling (HDH × WindSpeed).
- **Market-Aligned Resolution:** Smart meter intervals are aggregated to hourly energy totals (kWh/h), perfectly matching EPEX SPOT auction granularity while filtering localized stochastic switching noise.

3. Innovation & Originality

1. **Physical Solar Signature Disaggregation (Level 1b):** Unsurveyed households (165 of 410) lack PV metadata. We formulate a physics-based solar extractor that captures midday net load depression (R_midday = L̄_11–15 / L̄_01–05), duck-curve depth, and midday near-zero import frequency. A calibrated classifier achieves **0.8820 5-fold ROC-AUC** (92.3% precision) on surveyed households and resolves PV prosumer status across the entire portfolio.
2. **Asymmetric Risk Optimization via Quantile Duality (Level 3):** Standard deterministic forecasts (mean/median) inherently assume symmetric penalties and incur severe deficit costs. Because balancing market losses follow check/pinball loss, the cost-minimizing bidding volume is proven to be the critical quantile:
3. $$\tau^* = \frac{c_{\text{short}}}{c_{\text{short}} + c_{\text{long}}} = \frac{35}{35 + 15} = 0.70$$

τ = c_short / (c_short + c_long) = 35 / (35 + 15) = 0.70*

By fitting multi-quantile gradient boosted trees, we directly translate uncertainty into optimal bidding decisions.

4. Experimental Design & Numerical Performance

Evaluated on a strictly chronological holdout test set (119,541 hourly observations spanning Fall/Winter 2023–2024; zero data leakage):

Model Level	Model Architecture	MAE (kWh)	RMSE (kWh)	WAPE (%)	R^2 Score	Specific Cost (€/MWh)	Balancing Cost (€)	Cost Reduction
Level 0 Baseline	Seasonal Persistence (t−168 h)	0.7661	1.3925	54.25%	0.1701	€13.80	€2,329.63	Baseline
Level 0 Baseline	Regularized Ridge Regression	0.6261	1.0631	44.34%	0.5163	€11.28	€1,903.46	−18.29%
Level 1 Advanced	Global LightGBM Regressor	0.5754	1.0620	40.75%	0.5172	€11.44	€1,931.54	−17.09%
Level 1 Advanced	Stratified Prosumer Forecaster	0.5714	1.0585	40.46%	0.5204	€11.36	€1,917.06	−17.71%
Level 3 Optimal	**Cost-Optimal Quantile Bid ($\tau^* = 0.70$)**	—	—	—	—	**€10.08**	**€1,702.08**	**−26.94%**

- **Accuracy Gains:** Stratified LightGBM achieves a **13.79 percentage point reduction in WAPE** (54.25% → 40.46%) and triples explained variance (R^2: 0.17 → 0.52).
- **Uncertainty Calibration:** The 80% nominal prediction interval achieves **78.64% empirical coverage (PICP)** with a sharp mean width of 1.74 kWh/h.
- **Financial Impact:** Bidding at the cost-optimal quantile $\tau* = 0.70$ reduces total balancing market expenditure by **26.94% (€627.55 savings)** vs baseline, unlocking an extra **€261.81 (13.33%)** in financial savings compared to deterministic median bidding.

5. Uncertainty Diagnostics & Operational Value for E.ON

- **Where Uncertainty Concentrates:** Uncertainty interval widths expand by **3.3×** during freezing temperatures ($T < 0$ °C, 2.51 kWh/h) compared to mild conditions ($T > 20$ °C, 0.78 kWh/h) due to heat pump COP degradation and auxiliary resistive heater cycling. Prosumers exhibit pronounced midday uncertainty (solar cloud transients), whereas pure heat-pump households show nighttime heating variance.
- **Strategic Deployment:** E.ON can deploy this modular engine directly within energy trading desks: dynamically shifting bidding quantiles based on real-time price spreads and dispatching heat pump flexibility during solar noon troughs.

NOTE: Hackathon Team

Name: Gilbert Tcheugoue Djissitche

I sincerely apologize for being unable to attend the JARA-ENERGY Summer School 2026 in Eupen. I applied for my visa as soon as I was able to, but unfortunately, I received a message from the Embassy this morning informing me that my visa application has been forwarded to the Immigration Office (SPF Interior) for examination and decision. As a result, I have not yet received my visa and was unable to travel to Eupen in time for the Summer School.

I sincerely regret missing this valuable opportunity and would like to thank you for selecting me and for your understanding. I was very much looking forward to participating, learning from the speakers and other participants, and contributing to the discussions. I hope to have the opportunity to participate in a future JARA-ENERGY event.

Due to the visa situation mentioned above, I was unable to travel to Eupen and therefore participated in the hackathon remotely. As a result, I worked individually on the project from Cameroon rather than with an on-site team.